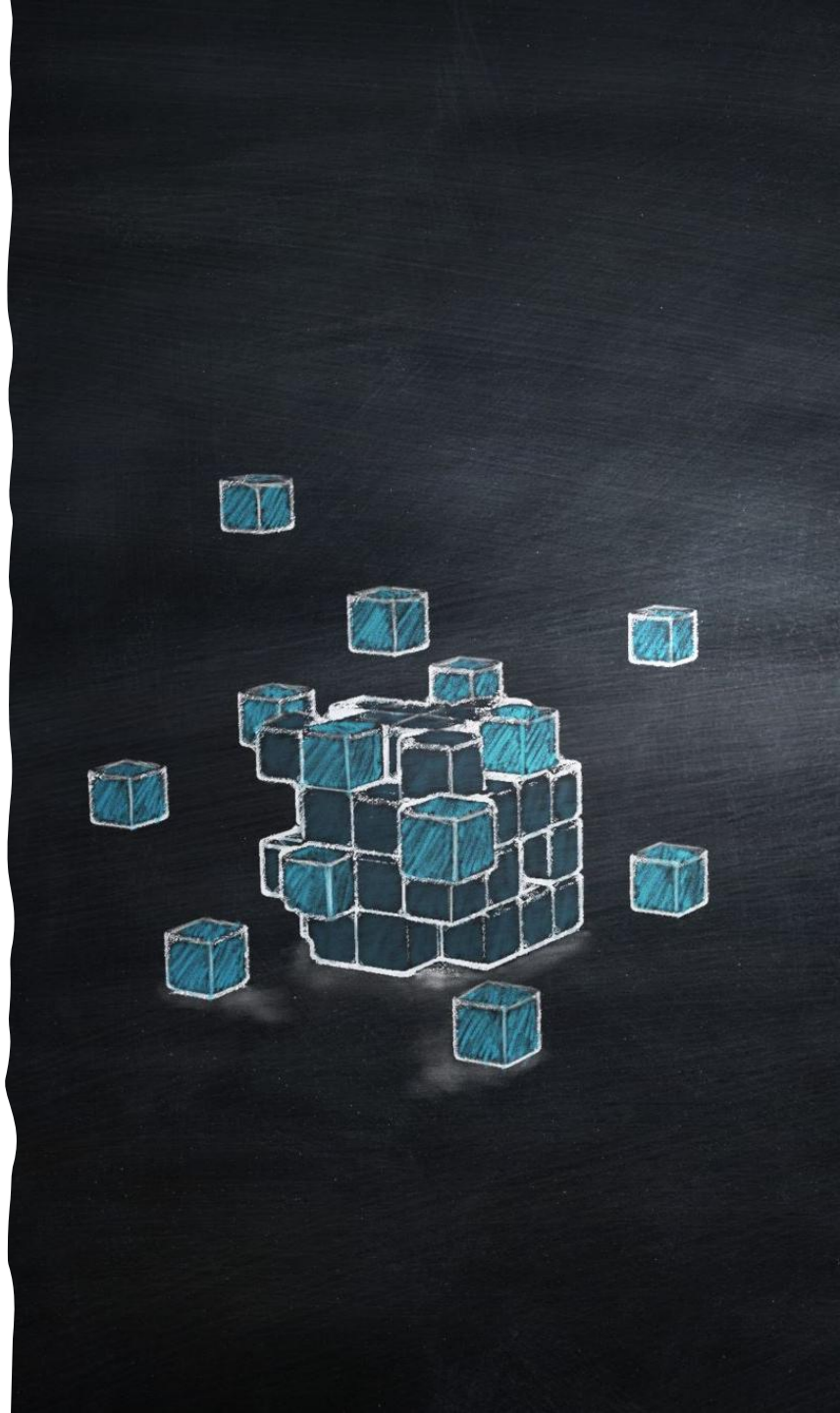
Sentence Transformers

understanding sentence embedding vectors and cosine similarity

Art Dewulf







Star 19,150

Search docs

GETTING STARTED

- Installation
- Quickstart
- Migration Guide

SENTENCE TRANSFORMER

- Usage
- Pretrained Models
- Speeding up Inference
- Training Overview
- Dataset Overview
- Loss Overview
- Training Examples

Sentence Transformers library

```python
from sentence_transformers import SentenceTransformer

model = SentenceTransformer("sentence-transformers/all-MiniLM-L6-v2")

# Two lists of sentences
sentences1 = [
    "The new movie is awesome",
    "The cat sits outside",
    "A man is playing guitar",
]

sentences2 = [
    "The dog plays in the garden",
    "The new movie is so great",
    "A woman watches TV",
]

# Compute embeddings for both lists
embeddings1 = model.encode(sentences1)
embeddings2 = model.encode(sentences2)

# Compute cosine similarities
similarities = model.similarity(embeddings1, embeddings2)
```

```
The new movie is awesome
- The dog plays in the garden    : 0.0543
- The new movie is so great      : 0.8939
- A woman watches TV             : -0.0502
The cat sits outside
- The dog plays in the garden    : 0.2838
- The new movie is so great      : -0.0029
- A woman watches TV             : 0.1310
A man is playing guitar
- The dog plays in the garden    : 0.2277
- The new movie is so great      : -0.0136
- A woman watches TV             : -0.0327
```

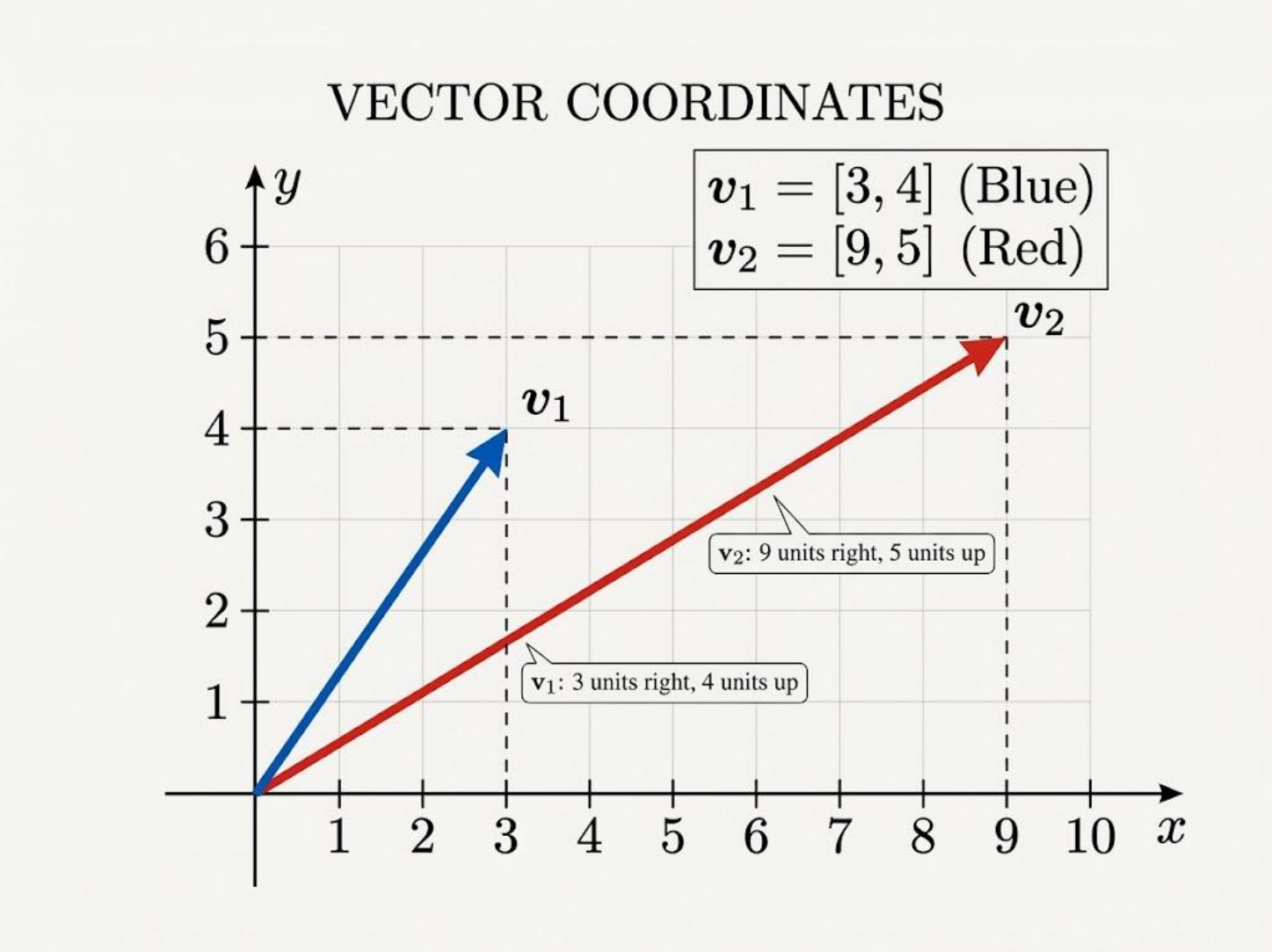


From transformers to sentence transformers

- Sentence Transformer models are based on more basic **transformer language models** (e.g. BERT, GPT)
- The values in the final layer of a BERT model (just before the word prediction layer) is list of numbers (an embedding vector) that captures much of the contextual meaning of the masked word
- These contextual embeddings of each word are then pooled (mostly averaged) into a **sentence embedding vector**
- Sentence Transformers models are further **trained** with triplets (sentence, similar sentence, different sentence) to produce sentence embedding vectors that reflect sentence similarity
- Such that **cosine similarity** between sentence embedding vectors reflects **semantic similarity** between the original sentences

Vectors and coordinates

- Vectors have a size and a direction
- A vector in a 2D space has 2 coordinates
- A vector in a 3D space has 3 coordinates
- A vector in a 768D space has 768 coordinates!
- In Python, a vector is represented as list of numbers (coordinates)

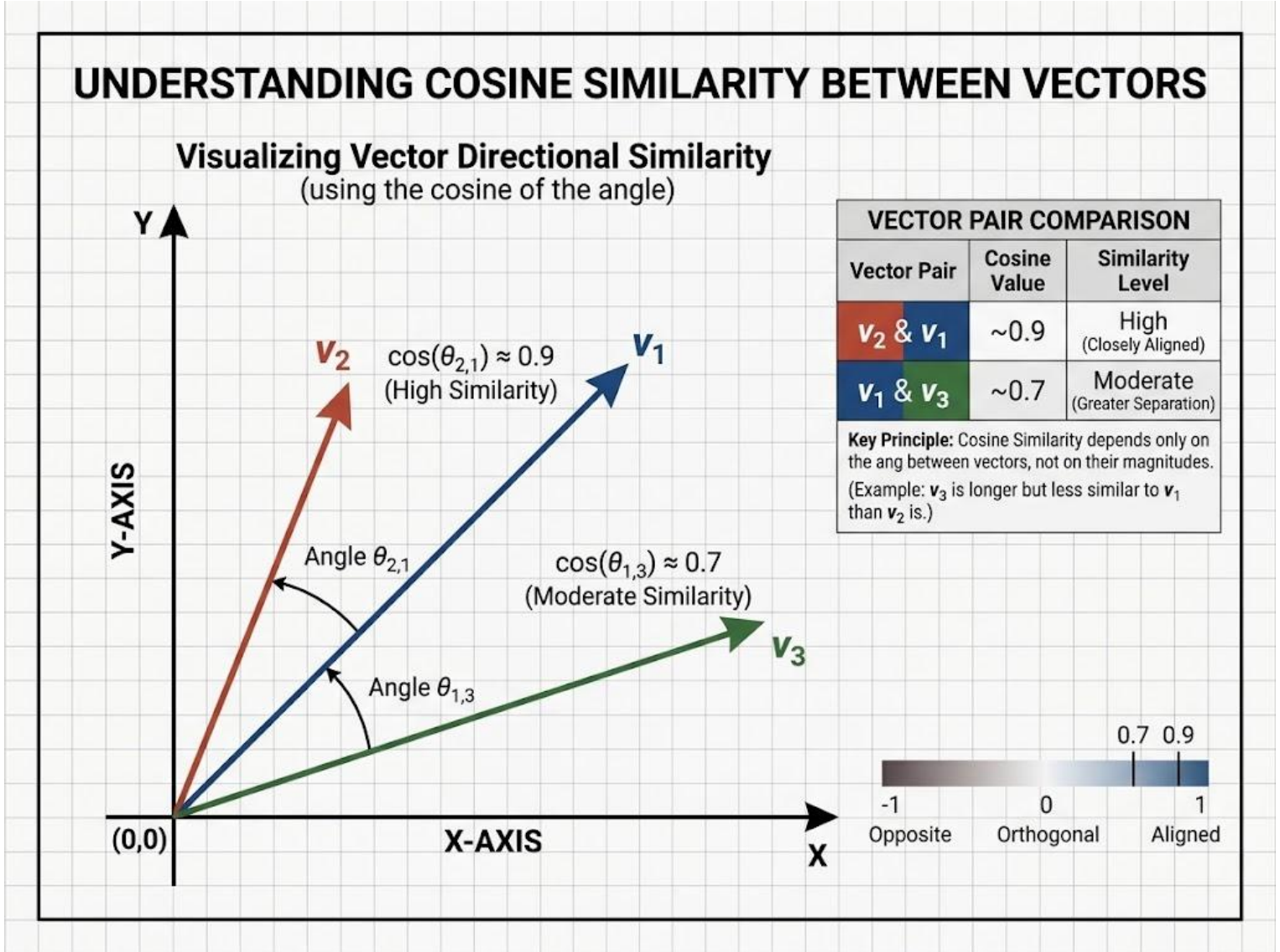


Cosine Similarity

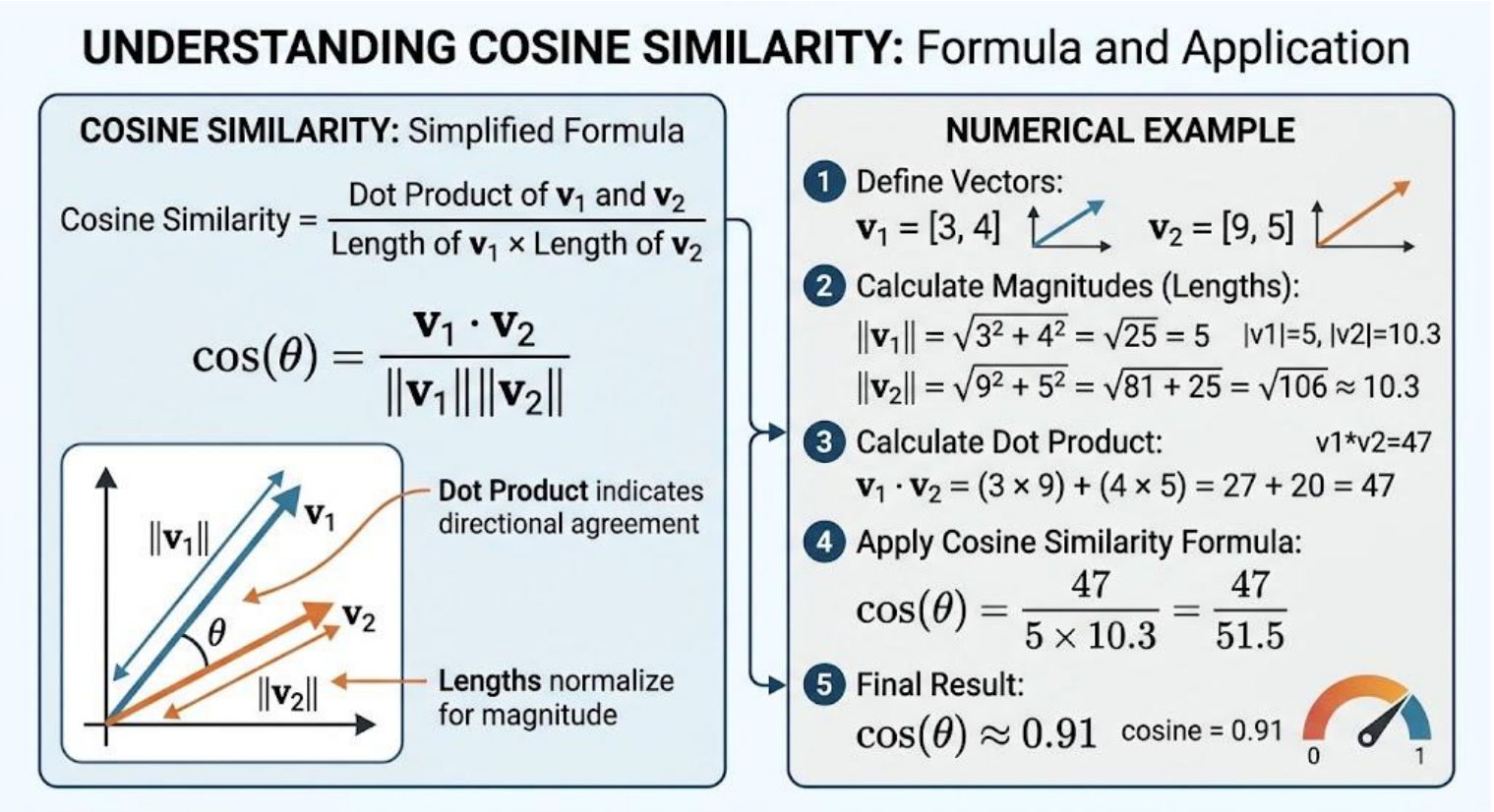
ranges from **1** (identical direction)

over **0** (orthogonal)

to **-1** (opposite direction)



Calculating cosine similarity: an example



Applications of cosine similarity between text embedding vectors

Retrieval	Deduplication	Keyword extraction	Sentiment analysis	Classification
• compare a query to a set of texts and list the most similar texts	• filter out near-identical texts using a high similarity threshold	• extract the most central words from a text by comparing the word embeddings with the sentence embedding	• comparing texts with embeddings of ‘positive’ versus ‘negative’	• create an anchor embedding for each category and assign documents to the category to which they are most similar

Application: topic modelling pipeline

Embed

- Use a sentence transformer model to generate text embedding vectors for the corpus

Reduce dimensionality

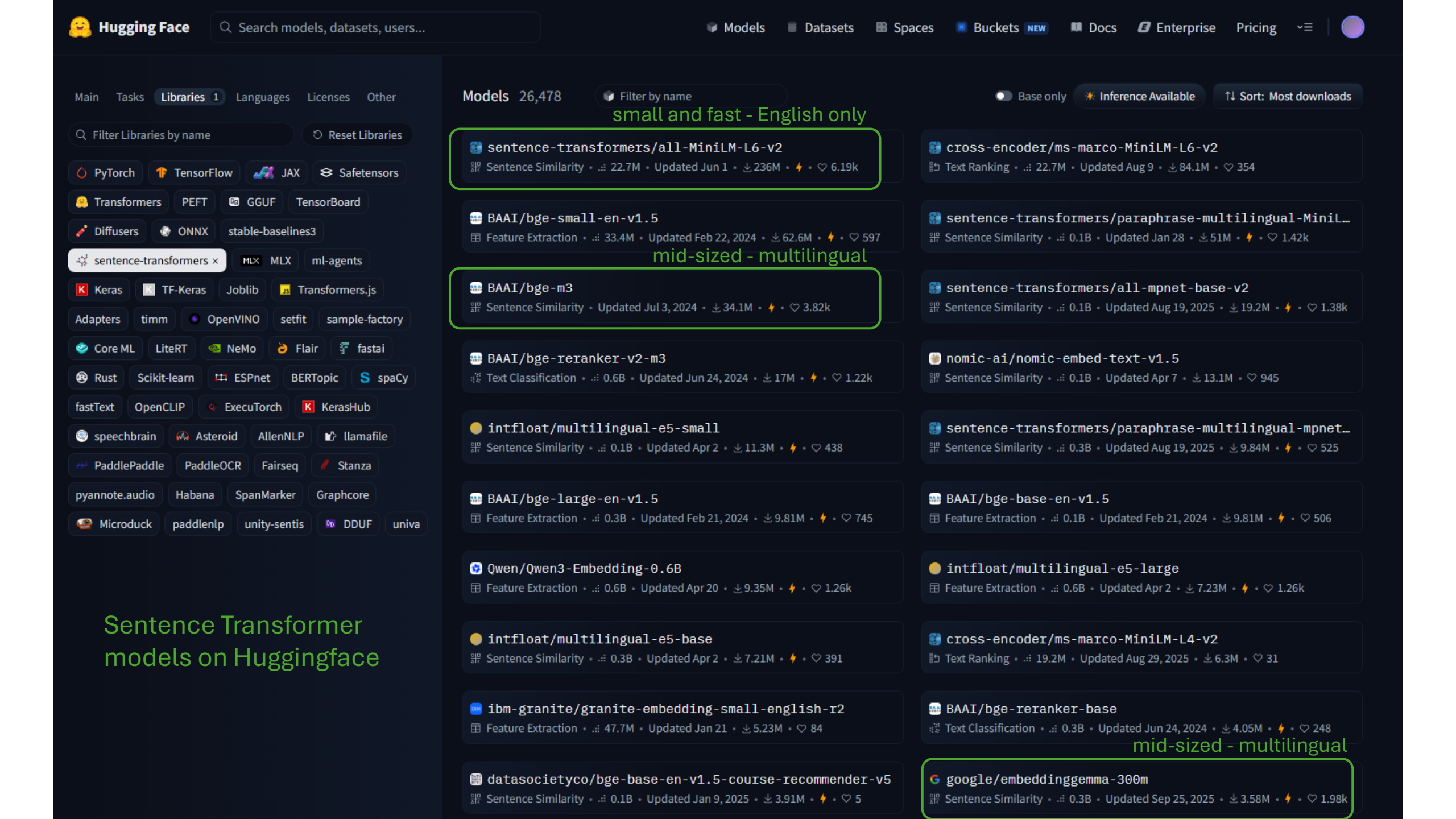
- Reduce the high-dimensional embedding vectors to low-dimensional vectors to allow clustering

Cluster

- Identify clusters of texts with similar meaning to create topics

Represent

- Represent topics by extracting typical keywords for each cluster of documents


Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Buckets NEW
Docs
Enterprise
Pricing
Main
Tasks
Libraries 1
Languages
Licenses
Other
Filter Libraries by name
Reset Libraries
PyTorch
TensorFlow
JAX
Safetensors
Transformers
PEFT
GGUF
TensorBoard
Diffusers
ONNX
stable-baselines3
sentence-transformers ×
MLX
ml-agents
Keras
TF-Keras
Joblib
Transformers.js
Adapters
timm
OpenVINO
setfit
sample-factory
Core ML
LiteRT
NeMo
Flair
fastai
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Scikit-learn
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Fairseq
Stanza
pyannote.audio
Habana
SpanMarker
Graphcore
Microduck
paddlenlp
unity-sentis
DDUF
univa
Sentence Transformer models on Huggingface
Models 26,478
Filter by name
Base only
Inference Available
Sort: Most downloads
small and fast - English only
sentence-transformers/all-MiniLM-L6-v2
Sentence Similarity • 22.7M • Updated Jun 1 • 236M • 6.19k
cross-encoder/ms-marco-MiniLM-L6-v2
Text Ranking • 22.7M • Updated Aug 9 • 84.1M • 354
BAAI/bge-small-en-v1.5
Feature Extraction • 33.4M • Updated Feb 22, 2024 • 62.6M • 597
sentence-transformers/paraphrase-multilingual-MiniL…
Sentence Similarity • 0.1B • Updated Jan 28 • 51M • 1.42k
mid-sized - multilingual
BAAI/bge-m3
Sentence Similarity • Updated Jul 3, 2024 • 34.1M • 3.82k
sentence-transformers/all-mpnet-base-v2
Sentence Similarity • 0.1B • Updated Aug 19, 2025 • 19.2M • 1.38k
BAAI/bge-reranker-v2-m3
Text Classification • 0.6B • Updated Jun 24, 2024 • 17M • 1.22k
nomic-ai/nomic-embed-text-v1.5
Sentence Similarity • 0.1B • Updated Apr 7 • 13.1M • 945
intfloat/multilingual-e5-small
Sentence Similarity • 0.1B • Updated Apr 2 • 11.3M • 438
sentence-transformers/paraphrase-multilingual-mpnet…
Sentence Similarity • 0.3B • Updated Aug 19, 2025 • 9.84M • 525
BAAI/bge-large-en-v1.5
Feature Extraction • 0.3B • Updated Feb 21, 2024 • 9.81M • 745
BAAI/bge-base-en-v1.5
Feature Extraction • 0.1B • Updated Feb 21, 2024 • 9.81M • 506
Qwen/Qwen3-Embedding-0.6B
Feature Extraction • 0.6B • Updated Apr 20 • 9.35M • 1.26k
intfloat/multilingual-e5-large
Feature Extraction • 0.6B • Updated Apr 2 • 7.23M • 1.26k
intfloat/multilingual-e5-base
Sentence Similarity • 0.3B • Updated Apr 2 • 7.21M • 391
cross-encoder/ms-marco-MiniLM-L4-v2
Text Ranking • 19.2M • Updated Aug 29, 2025 • 6.3M • 31
ibm-granite/granite-embedding-small-english-r2
Feature Extraction • 47.7M • Updated Jan 21 • 5.23M • 84
BAAI/bge-reranker-base
Text Classification • 0.3B • Updated Jun 24, 2024 • 4.05M • 248
mid-sized - multilingual
datasocietyco/bge-base-en-v1.5-course-recommender-v5
Sentence Similarity • 0.1B • Updated Jan 9, 2025 • 3.91M • 5
google/embeddinggemma-300m
Sentence Similarity • 0.3B • Updated Sep 25, 2025 • 3.58M • 1.98k